Tab 2

B.Sc. (Hons.) IT (Data Science)

No.	Question	CO	PO	BTL
Unit-1: Fundamental of Algorithms				
1)	What is an Algorithm? Explain its characteristics and rules with an example.	CO1	1,2,7	L2
2)	Explain Sequence, Branching and Looping algorithms with suitable examples.	CO1	1,2,7	L2
3)	What is a Flowchart? Explain various flowchart symbols used in problem-solving.	CO1	1,2,7	L2
4)	State the general rules for designing a flowchart.	CO1	1,2,7	L2
5)	Differentiate between Algorithm and Flowchart.	CO1	1,2,7	L2
6)	Develop an algorithm and flowchart to find the largest of three numbers.	CO1	1,2,7	L3
Unit-2: Overview of C				
1)	Discuss the history, importance and features of C language.	CO2	1,2,5,7	L2
2)	Explain the basic structure of a C program and the steps involved in program execution.	CO2	1,2,5,7	L2
3)	Explain C Tokens with suitable examples.	CO2	1,2,5,7	L2
4)	What are variables? Explain variable declaration rules and primary data types in C.	CO2	1,2,5,7	L2
5)	Explain constants and symbolic constants with examples.	CO2	1,2,5,7	L2
6)	Discuss User Defined Type Declaration (typedef) with example.	CO2	1,2,5,7	L2
Unit-3: Operators and Expressions				
1)	What is an operator? Explain Arithmetic, Relational and Assignment operators with examples.	CO3	1,2,5,7	L2

2)	Explain Logical, Increment and Decrement operators with examples.	CO3	1,2,5,7	L2
3)	Discuss Bitwise and Special operators in C language.	CO3	1,2,5,7	L2
4)	Explain Conditional (Ternary) Operator with suitable example.	CO3	1,2,5,7	L2
5)	Explain operator precedence, associativity and type conversion in C language.	CO3	1,2,5,7	L3
6)	Write short notes on input/output functions printf() and scanf() with examples.	CO3	1,2,5,7	L2
Unit-4: Decision Making and Branching				
1)	Explain simple if and if-else statements with suitable examples.	CO4	1,2,5,7	L2
2)	Explain Nested if-else and else-if ladder statements with examples.	CO4	1,2,5,7	L2
3)	Explain switch-case statement with a suitable example.	CO4	1,2,5,7	L2
4)	Discuss goto, break and continue statements with examples.	CO4	1,2,5,7	L2
5)	Compare while, do-while and for loops with suitable examples.	CO4	1,2,5,7	L3
6)	Write a C program using decision-making and looping statements to display student grades.	CO4	1,2,5,7	L3
Unit-5: Arrays and Strings				
1)	What is an array? Explain types of arrays with examples.	CO5	1,2,5,7	L2
2)	Discuss compile-time and run-time initialization of one-dimensional arrays.	CO5	1,2,5,7	L2
3)	Explain Two-Dimensional and Multi-Dimensional arrays with examples.	CO5	1,2,5,7	L2

4)	What is a Character Array? Explain declaration and initialization of string variables.	CO5	1,2,5,7	L2
5)	Explain getchar(), gets(), putchar() and puts() functions with examples.	CO5	1,2,5,7	L2
6)	Explain string handling functions (strlen(), strcpy(), strcat(), strcmp()) with examples.	CO5	1,2,5,7	L3